Selected Experience and Rare-Event Risk: Endogenous Observation, Subjective Transformation, and Reputation under Hidden Counterfactuals

Abstract

Risky decisions about rare events—insurance against uncommon losses, credit for rare defaults, low-frequency professional services—often rest on histories that are selectively generated: an outcome enters the record only when someone participated and observed it. This paper studies how such endogenous observation changes experienced risk, how subjective probability transformation and reputation act on that selected experience, and what can and cannot be inferred about underlying efficacy when counterfactuals are hidden. It separates four objects: the objective probability, the experienced frequency on the selected record, a transformed subjective probability, and a price-implied index from a reduced-form quote rule. A standard selection identity gives the ratio of experienced to objective risk as one plus a scaled covariance between state risk and observation; in the calibrated simulation the ratio is approximately 1.71 and does not depend on baseline rarity, while the absolute gap shrinks with rarity. Rarity instead changes the composition of the quote: the relative distortion of the reputation-augmented quote rises from 1.12 to 11.61 as objective expected value falls, through probability weighting and an additive reputation stock that is decoupled from current hazard. In simulated rare regimes, zero-efficacy selection matches the mean apparent performance of positive efficacy under exogenous observation. A weather benchmark with zero causal efficacy by construction—rainmaking pseudo-events over 87,664 region-days—shows apparent success of 0.853 and model reputation premia alongside negative price–success differences.

Keywords: decisions from experience; endogenous observation; rare events; probability weighting; reputation; selection

JEL classification: D81; D82; D83; D84

1. Introduction

Many economically consequential transactions price the occurrence of rare, partly controllable events: surgical outcomes for uncommon conditions, catastrophe insurance, drought interventions, disaster-recovery contracts, and the advertised efficacy of low-frequency professional services. These transactions share three structural features that standard pricing analysis usually treats separately. The event is rare, so both buyers and sellers learn from sparse positive outcomes. The counterfactual is hidden, because the would-be outcome without intervention is not observed for accepted cases. And observation is endogenous: which interventions occur—and which are recorded—depends on the same risk state that generates outcomes. Humans have long attributed causal efficacy to coincident outcomes under exactly these conditions (Skinner, 1948), and forecast evaluation has long required scoring performance relative to the difficulty of the task (Murphy, 1993). The question motivating this paper is deceptively simple: when these three features hold, what does a posted quote actually encode? The answer requires distinguishing four objects that are often conflated. The objective probability p_t is the data-generating risk state. The experienced probability $\hat{p}_{exp,t}$ is the frequency of apparent success in the cases that were actually observed. The subjective probability $\hat{p}_{subj,t}$ applies a transformation—Bayesian updating, selective memory, or a parameterized probability-weighting scenario—to that experienced frequency. The fourth object, $\hat{q}_{price,t}$, is a price-implied index: the value an expected-value benchmark would have to place on success to rationalize the quoted price. It is deliberately not called a probability: under reputation-mediated pricing it includes an additive stock term and can exceed one, so it is a

composite index rather than a quantity constrained to $[0,1]$. Throughout, "price" and "pricing" refer to this model-implied quote formation under a reduced-form pricing rule, not to market-clearing equilibrium prices. Under exogenous observation the objective and experienced layers coincide in expectation when defined on the same support; subjective transformation and reputation can still separate the later layers. Endogenous observation introduces an additional composition gap between the objective and experienced probabilities, and a price can be simultaneously inflated relative to objective risk and deflated relative to experienced frequency.

A second question follows: what can an outsider infer about true efficacy from observed prices? We show that endogenous selection can create a mean-performance equivalence—in the simulated rare regimes, environments with exactly zero true efficacy generate the same MEAN apparent performance as positive efficacy under exogenous observation—so mean apparent performance alone cannot distinguish the two environments without auxiliary structure. The equivalence is first-moment only: it does not equate the joint distribution of outcomes, prices, provider identities, or histories. Model-implied quotes transmit a mixture of baseline hazard, selection, transformation, and reputation rather than a clean probability signal, so observed quote data do not automatically solve the problem either. The paper contains no human-subject or market-transaction data: its claims concern what model-implied quotes can encode, which is a statement about information structure rather than estimated behavior.

The paper makes three contributions, each a result rather than a combination of components. First, it formalizes the four-layer structure and proves a matched-support decomposition: the experienced-frequency gap splits into a selection/composition term and a same-support record term, and under the model's multiplicative hazard the selection ratio equals one plus a scaled covariance between state risk and the observation weight, analytically independent of baseline

rarity, with its sign set by that covariance (Lemma 1)—the simulated ratio near 1.71 is the positive-selection calibration, not a universal constant. Second, it separates probability distortion from price-composition distortion: a generic simulation varying rarity, observation rules, learning rules, and pricing models shows that rarity leaves the frequency structure essentially unchanged while the relative composition of modeled prices shifts through two non-scaling channels—a nonlinear probability-transformation term and an additive, history-dependent reputation term—both of which can grow relative to the objective-EV benchmark as that benchmark shrinks. In the simulated finite regimes the transformed-frequency term is quantitatively the larger share of the total relative distortion, which rises from 1.12 to 11.61 while the absolute price deviation stays flat; reputation's distinct role is a stock–flow decoupling in which reputation premia coexist with zero or negative contemporaneous price-success differences. The simulation also establishes a first-moment equivalence between zero-efficacy selection and positive efficacy. Third, it benchmarks the mechanism against a real stochastic process with known-zero causal efficacy: rainmaking pseudo-events over 87,664 archived region-days of observed rainfall and 2595 synthetic pseudo-events, where every causal efficacy attribution is false by construction and the selection, attribution, and reputation mechanisms still produce elevated apparent performance.

Two clarifications frame the contribution. The selection mechanism is GENERAL—it operates for common events too; what is specifically rare-event about this paper is the economic scaling: as objective expected value shrinks, non-scaling price components become relatively dominant and successful outcomes become sparser, while inference from raw frequencies weakens. The ingredients are established and are cited as such. Endogenous sampling biases experience-based estimates because observation depends on the learner's own state or choices (Denrell & March,

2001; Fiedler, 2000; Denrell, 2005), and the bias survives Bayesian updating (Le Mens & Denrell, 2011). Selected-sample means differ from population means by a covariance term (Heckman, 1979), and selective acceptance distorts provider report cards (Dranove et al., 2003; Werner & Asch, 2005). Rare events are underweighted when experienced through sampling rather than described (Barron & Erev, 2003; Hertwig et al., 2004; Hertwig & Erev, 2009; Wulff et al., 2018), stated probabilities differ from experienced ones (Ungemach et al., 2009), and low probabilities are hard to use in decisions at all (Camerer & Kunreuther, 1989; Kunreuther et al., 2001). Berk–Nash equilibrium (Esponda & Pouzo, 2016), analogy-based expectations (Jehiel, 2018), and laboratory evidence (Esponda & Vespa, 2018) show that endogenous feedback can make mislearning persistent. Probability weighting supplies the transformation layer (Tversky & Kahneman, 1992; Prelec, 1998; Gonzalez & Wu, 1999). The selection ratio below is a standard selection identity applied to the observation of a rare binary event, and the divergence of the transformed relative term is a known property of the Prelec function. The incremental contribution is the combination: an accounting decomposition of a reduced-form quote into objective, experienced, subjective, and reputation layers; the comparative static that separates rarity-invariant relative distortion from shrinking absolute distortion; a simulated stock–flow decoupling of reputation from current hazard; a first-moment equivalence between selection and positive efficacy; and a benchmark in which causal efficacy is zero by construction.

Section 2 defines the four probability objects. Section 3 states the theory, including the analytic selection-ratio lemma. Section 4 describes the generic simulation. Section 5 introduces the zero-effect environmental benchmark. Section 6 reports results. Section 7 discusses what prices can and cannot reveal, candidate application domains, and limitations. Section 8 concludes.

2. Conceptual framework

Figure 1 represents the data-generating system shared by the generic simulation and the environmental benchmark. A risk state enters demand, provider acceptance, valuation, attribution of outcomes, and reputation, while the intervention itself may or may not change the outcome. In the benchmark environment the causal arrow is deleted by construction; in the generic simulation its strength is a parameter swept from zero to positive values.

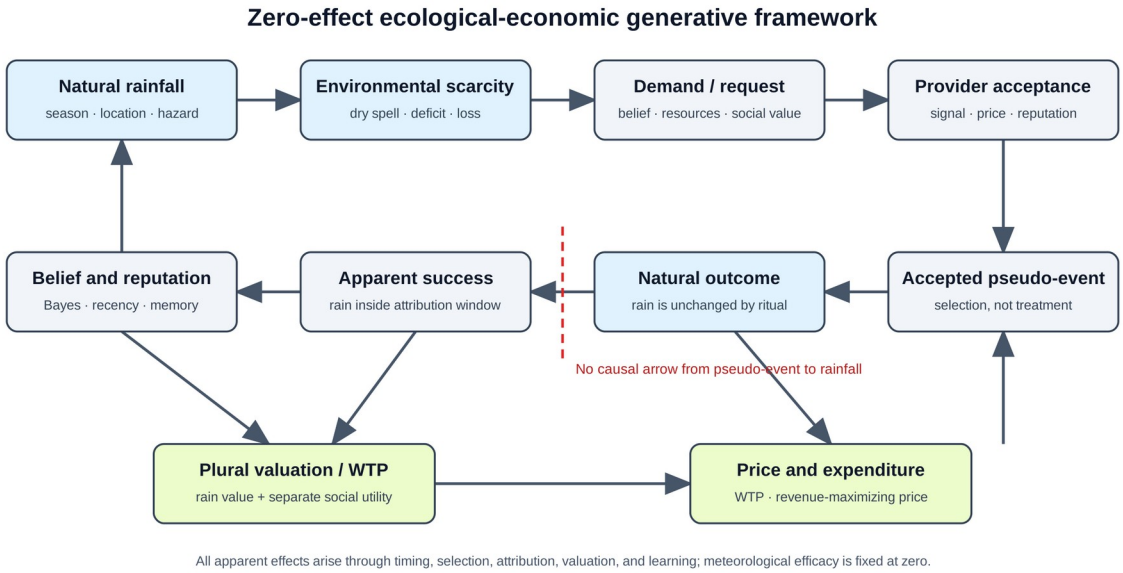


Fig. 1 Conceptual framework. Baseline environmental hazard enters demand, provider acceptance, valuation, attribution, and reputation, while the pseudo-intervention never changes rainfall

2.1. Four probability layers

For a case i in period t, let $S_{it} \in \{0,1\}$ indicate the risk state (e.g., high hazard), D_{it} the demand decision, A_{it} the provider's acceptance decision, O_{it} whether the case enters the observed record, and Y_{it} the attributed outcome. Four probability objects are defined on this structure:

$$p_t = P(Y_{it}=1 \mid S_{it}),\ \hat{p}_{exp,t} = \frac{\sum_i Y_{it} O_{it}}{\sum_i O_{it}},\ \hat{p}_{subj,t} = w(\hat{p}_{exp,t}),\ \hat{q}_{price,t} = \frac{P_t}{\alpha v},$$

where $w(\cdot)$ is a transformation (Bayesian updating, selective memory, or a probability-weighting function), P_t is the quote produced by a pricing rule (no market clearing, bargaining, or equilibrium demand is modeled), and v is the value of a successful outcome scaled by the pricing coefficient α. The first three objects are probabilities. The fourth is a price-implied index: under an additive reputation rule $P_t = \alpha \hat{w}_t v + \gamma R_t$ it equals $\hat{w}_t + \gamma R_t/(\alpha v)$, a composite that is not constrained to $[0,1]$ and is not interpretable as a probability in general. Objective success probability p_t and the success probability for accepted cases $P(Y=1 \mid A=1)$ differ whenever acceptance correlates with the risk state; experienced frequency adds the observation filter O_{it}; and the price-implied index absorbs reputation, margins, and the provider's selectivity.

The quote rule itself has a minimal reduced-form interpretation. If an agent quotes a share of perceived gross surplus, and perceived benefit is the transformed probability times the value of a success, then the probability part of the quote is $\alpha \hat{w}_t v$, where α is the share of perceived gross surplus the quote recovers. Reputation enters as a separable trust/amenity component γR_t, where γ is the exchange rate between accumulated reputation and quote units. The reputation-mediated rule $P_t = \alpha \hat{w}_t v + \gamma R_t$ is therefore a reduced-form mapping from perceived gross benefit plus a separable reputation component to a posted quote; it makes no equilibrium, bargaining, or welfare-optimality claim, and the objective-EV benchmark used below is normative only in the limited sense of objective expected value under the model, not a welfare optimum.

2.2. Reputation as a stock, performance as a flow

Provider reputation R_{jt} is a decaying success rate over the provider's own attributed outcomes before t:

$$R_{jt} = \frac{r_0 + \sum_{\tau<t}(1-\rho)\rho^{t-1-\tau} O_\tau Y_\tau 1[j_\tau = j]}{n_0 + \sum_{\tau<t}(1-\rho)\rho^{t-1-\tau} O_\tau 1[j_\tau = j]},$$

with decay ρ and prior stocks r_0, n_0 (Section 4),

while contemporaneous apparent success is a flow driven by the current risk state and the current case mix. Because the stock and the flow answer different questions, a reputation-mediated price premium can coexist with weak, zero, or negative mean contemporaneous price-success differences—a decoupling that the benchmark environment makes measurable. This stock–flow separation is the framework's central testable implication: where prices encode reputation, the contemporaneous correlation between what was paid and what was delivered need not be positive.

2.3. The mean-performance equivalence problem

An outside observer sees prices, attributed outcomes, and the accepted case mix, but not the rejected cases or the counterfactual. If endogenous selection concentrates interventions in high-risk states, apparent success rises even when efficacy is exactly zero. Formally, let δ denote true efficacy and σ the strength of selection. Mean-performance equivalence holds when $(\delta = 0, \sigma > 0)$ under endogenous observation produces the same MEAN apparent success as $(\delta > 0, \sigma = 0)$ under exogenous observation. The equivalence is deliberately first-moment: it makes no claim about the joint distribution of outcomes, prices, provider identities, event timing, or reputation trajectories, which can differ substantially across the two environments. Section 6 quantifies the equivalent δ implied by selection alone.

3. Theory

Propositions P1–P6 organize the framework; P2 is split into a supported form (P2a), an explicitly historical claim that failed (P2b), and the corrected comparative-statics statement (P2c). Lemma 1 supplies the analytic core of P2a. Table 1 classifies each by evidentiary status: demonstrated in the generic simulation, demonstrated in the zero-effect benchmark, or a theoretical implication not directly tested.

Table 1. Proposition classification.

Proposition	Classification	Basis
P1 Risk-correlated selection shifts experienced frequency; the sign is set by Cov(m, s)	Demonstrated (generic simulation and known-zero benchmark, positive-covariance calibration); derived analytically (Lemma 1)	selection ratio = 1 + Cov(m,s)/(E[m]E[s]); positive covariance yields selection_gap > 0 under endogenous observation at zero efficacy; adjusted hazard attenuation in the rainfall benchmark
P2a Selection shifts experienced frequency proportionally; positive-covariance calibration	Demonstrated (generic simulation); derived analytically (Lemma 1)	selection_ratio = p_ref/p_obj ≈ 1.71, analytically independent of baseline rarity under multiplicative state-dependent selection; ≈ 1.0 under exogenous observation; record term B_abs ≈ 0 on the observed support
P2b Original hypothesis: rarity amplifies the absolute selection gap	NOT SUPPORTED (negative result)	absolute selection gap falls with rarity (≈ 0.71 × p_obj); the original rarity-amplification hypothesis for the frequency gap was driven by scaling of the composition effect with p, not by a rarity mechanism
P2c Rarity changes the relative composition of modeled prices through nonlinear transformation and non-scaling reputation	Demonstrated (generic simulation); transformed term divergent analytically (Lemma 2)	D_rel(Model D) rises with rarity through both a nonlinear transformed-frequency part—the larger share in every simulated regime, asymptotically unbounded under Prelec weighting—and an additive reputation part whose share rises with rarity
P3 Weighting moves the subjective layer relative to experience	Demonstrated (generic simulation)	B_subj differs across learner rules; selective memory exceeds experienced frequency

Proposition	Classification	Basis
P4 Model-implied quotes mix hazard and transformation	Demonstrated (generic simulation)	B_price and D_EV differ across pricing models A–D
P5 Reputation decouples price from contemporaneous performance	Demonstrated (generic simulation; sharper in the known-zero benchmark)	Reputation premia coexist with small or negative contemporaneous price-success differences
P6 Mean-performance equivalence limits inference from raw performance	Demonstrated (generic simulation; first-moment only)	Zero efficacy under endogenous observation reproduces the MEAN apparent success of positive efficacy under exogenous observation in the simulated rare regimes (first-moment only; regime-dependent, no solution in the common regime)

P1 (Risk-correlated selection shifts experienced frequency; the sign is set by the covariance). If observation weights covary positively with the state risk multiplier, $E[p_t|O_t=1]>E[p_t]$ for the selected record even when efficacy is zero; negative covariance produces the opposite sign. This is the selection-on-gains channel of Heckman (1979) applied to experienced frequencies, generalized by Lemma 1 to an arbitrary state distribution: the composition shift is $Cov(m,s)/(E[m]E[s])$, and in the calibrated simulation the covariance is positive, converting baseline hazard into apparent performance.

Lemma 1 (Selection ratio under multiplicative state risk and state-dependent observation). Let the risk state be a random variable X with multiplier $m(X)>0$, and write the objective probability as $p(X)=p_0 m(X)$ where valid. Let the observation weight $s(X)\geq 0$—the probability that a case in state X enters the observed record—be independent of the baseline rarity scale p_0. Then population risk is $E[p]=p_0 E[m(X)]$, the selected-support risk is $p_{ref}=p_0 E[m(X)s(X)]/E[s(X)]$, and the selection ratio is

$$R=\frac{p_{ref}}{E[p]}=\frac{E[m(X)s(X)]}{E[m(X)]E[s(X)]}=1+\frac{Cov(m(X),s(X))}{E[m(X)]E[s(X)]},$$

which is independent of the baseline p_0: the rarity scale cancels in the ratio, and the absolute selection gap $p_{ref}-E[p]=p_0 E[m](R-1)$ is proportional to p_0. The identity also determines the

SIGN of the shift: positive $Cov(m, s)$ inflates the experienced frequency, zero covariance leaves the composition unchanged, and negative covariance deflates it—so risk-correlated selection need not inflate apparent success in general. The condition fails if observation responds to the absolute level $p(X)$ rather than to the state or relative hazard, in which case s is not independent of p_0.

The simulation's two-state process is the special case $X \in \{L, H\}$ with prevalence $\pi = P(X = H)$, multipliers $m_L < m_H$, mean multiplier $\mu = \pi m_H + (1 - \pi) m_L$, and state-dependent observation probabilities q_L, q_H (demand depends on the state; acceptance responds to the relative hazard $p_t / p_0 - 1 = m(S_t) - 1$). With $q_L = q_{D,L} q_{A,L}$ and $q_H = q_{D,H} q_{A,H}$,

$$R = \frac{\pi m_H q_H + (1 - \pi) m_L q_L}{\mu(\pi q_H + (1 - \pi) q_L)},$$

which evaluates to $R \approx 1.7115$ under the simulation's positive selection parameters; the simulated ratios (1.708–1.713) match within Monte Carlo error. The value 1.71 is parameter-specific, not universal, and the calibrated case is the positive-covariance example of the general identity.

P2a (Selection shifts experienced frequency proportionally; the simulation calibrates the positive-selection case). Holding selection strength fixed, the selection ratio is constant across rarity regimes by Lemma 1, so the absolute selection gap shrinks with rarity at the same rate as p itself; the calibrated simulation is the positive-covariance case. Moreover, the record term $E[Y \mid O = 1] - E[p_t \mid O = 1]$ is near zero at zero true efficacy: the shift is a composition effect on the selected support, not bias in the record itself.

P2b (Original hypothesis—not supported: rarity amplifies the absolute selection gap). The earlier formulation of this project predicted that the experienced-frequency gap would grow

monotonically with rarity. The matched-estimand decomposition rejects it: the absolute gap falls as fast as p and the ratio is flat. Reported as a negative result.

Lemma 2 (Unboundedness of the transformed relative term under Prelec weighting). Under the assumed one-parameter Prelec weighting $w(p)=\exp¿$ with $0<\alpha<1$, write $L=\ln(1/p)$. Then $w(p)/p=\exp(L-L^{\alpha})\to\infty$ as $p\to 0$, because $L-L^{\alpha}=L(1-L^{\alpha-1})\to\infty$ for $\alpha<1$. Hence $(w(p)-p)/p\to\infty$ as well: the transformed-frequency relative term is itself unbounded in rarity —it is not a bounded component. The finite simulation grid therefore compares two divergent terms at finite p, and which component is larger is a calibration fact, not an asymptotic one.

P2c (Rarity changes the relative composition of modeled prices through nonlinear transformation and non-scaling reputation). For a price $P=\alpha w^{¿}V+\gamma R$, $(P-\alpha pV)/(\alpha pV)=(w^{¿}-p)/p+\gamma R/(\alpha pV)$. Under the assumed Prelec weighting the first term diverges as $p\to 0$ (Lemma 2), and the second diverges provided the reputation stock R does not itself vanish proportionally with p—the condition the simulation parameterizes. Both components therefore fail to scale with objective expected value: rarity changes which components dominate the price signal relative to the objective-EV benchmark, not how large the frequency distortion is in absolute terms. In the simulated finite regimes the transformed-frequency term is the larger of the two; reputation's distinct contribution is the stock–flow decoupling of P5.

P3 (Nonlinear transformation moves the subjective layer relative to experience). For rare events, an inverse-S probability-weighting function $w(p)$ overweights p (Prelec, 1998; Tversky & Kahneman, 1992), which can move $p^{¿}_{subj}$ away from the experienced frequency in either direction. The Prelec transformation used here is one parameterized scenario, not an estimated behavioral law and not the same mechanism as the description–experience underweighting

documented for learned frequencies (Hertwig et al., 2004; Ungemach et al., 2009; Wulff et al., 2018); Bayesian updating, recency weighting, and selective memory are alternative transformations that the simulation compares separately.

P4 (Model-implied quotes mix hazard and transformation). Under expected-value pricing with linear transformation, the price-implied index tracks p^{ι}_{subj}. Under reputation-mediated pricing it additionally loads on the reputation stock, so q^{ι}_{price} is no longer interpretable as a probability at all but a composite of hazard, transformed frequency, and accumulated attribution. Misspecified-feedback equilibria make the persistence of this composite systematic rather than transient (Esponda & Pouzo, 2016; Jehiel, 2018; Esponda & Vespa, 2018), and bounded learning on endogenous variables can degrade inference further (Shea, 2015).

P5 (Reputation decouples price from performance). Because reputation is a stock, periods of high current hazard with undelivered outcomes can coincide with high prices built on earlier attributed successes. The contemporaneous price-success association can therefore be zero or negative while a reputation premium is strongly positive.

P6 (Mean-performance equivalence limits inference from raw performance). Endogenous selection under zero efficacy can reproduce—and in some regimes exceed—the MEAN apparent performance generated by positive efficacy under exogenous observation: mean apparent performance alone cannot distinguish the two environments in several simulated rarity regimes. The equivalence is first-moment and regime-dependent: it does not equate distributions, and in the simulated common regime no efficacy in the scanned grid reproduces the endogenous apparent success, so it is not universal. Observed quote data do not automatically solve the problem either, because in the model quotes themselves load on transformed beliefs and reputation—a caution, not a proved identification theorem. Within the present framework,

separating selection from efficacy can be aided by additional information such as rejected-case disclosure, an exogenous observation rule, pre-event risk information, or a known-zero benchmark; other identification strategies may also be possible.

4. Generic simulation

The generic simulation implements the framework as a discrete-time stochastic system. A two-state risk process S_t persists with transition probability 0.97 and spends 30% of periods in the high-hazard state, in which the success probability is multiplied by 2.0 relative to baseline (by 0.6 in the low-hazard state). Four rarity regimes set the baseline success probability p to 0.30 (common), 0.10 (moderately rare), 0.03 (rare), and 0.01 (very rare). Demand increases in the current risk state; provider acceptance increases in the relative hazard $p_t/\bar{p} - 1$ with provider-specific selectivity; accepted cases enter the observed record (endogenous rule), while an exogenous comparator requests interventions at the same marginal rate and accepts at a constant rate, so demand and acceptance are both independent of the state. Every learner state, reputation stock, and quote at period t is computed from the observed record through $t-1$ only, so no quote conditions on the outcome it prices. The naive learner reports the raw experienced frequency $\hat{p}_{exp,t-1} = \sum_{\tau<t} Y_\tau O_\tau / \sum_{\tau<t} O_\tau$; before the first observation, where this ratio is undefined, it is initialized at the pre-specified prior mean $\bar{p}$ (an initialization, not smoothing). The subjective-learning variants are Bayesian updating with a Beta prior of mean $\bar{p}$ and strength 2.0, exponential recency weighting with decay 0.995, and selective memory that weights attributed successes at 1.0 and failures at 0.35; the recency, selective-memory, and reputation stocks start from a prior weight equivalent to 20 attributed outcomes at mean 0.5. Subjective probabilities for models C and D apply a Prelec weighting function with $\alpha = 0.65$ to the Bayesian estimate, a

parameterized scenario drawn from the literature, not an estimate from human subjects or market data in this paper. Prices are computed under four pricing rules—each produces a model-implied quote, and none models market clearing: A, expected value on the objective probability; B, expected value on the raw experienced frequency $\hat{p}_{exp,t-1}$; C, a probability-weighted quote; and D, a reputation-mediated rule that adds a fraction 0.5 of the provider's reputation stock to the quote base. Reputation decays at $\rho = 0.995$ per period from prior stocks $r_0 = 0.05$, $n_0 = 0.10$, and it enters the quote in periods whose case is observed for that provider; in other periods the model D quote equals the model C quote. Each cell uses 2,000 independent replications of a $4,000$-period horizon after burn-in; true efficacy δ is swept over $\{0, 0.02, 0.05, 0.10\}$.

Metrics are defined per layer and averaged over replications with Monte Carlo error bands. The estimands are matched by construction (estimand audit in the project documentation): $p_{ref} = E[p_t \mid O_t = 1]$ is the objective probability on the observed support; the selection gap $p_{ref} - E[p_t]$ and selection ratio $p_{ref} / E[p_t]$ measure the composition shift produced by endogenous observation; the matched record bias $B_{abs} = \hat{p}_{exp} - p_{ref}$ and $B_{rel} = B_{abs} / p_{ref}$ compare the experienced frequency to its own estimand; under exogenous observation demand and acceptance are both state-independent, so $p_{ref} = E[p_t]$ and all three selection quantities are zero. the subjective and price layers are measured against the objective probability, $B_{subj} = \hat{p}_{subj} - p$ and $B_{price} = \hat{q}_{price} - p$, averaged over all periods (increments between adjacent layers follow by differencing the layer means reported in Table 2 and Table S8, and are not labeled B); $D_{EV} = price - \alpha v p$ is the deviation from expected value; and D_{rel} is the period mean of $D_{EV} / (\alpha v p)$, the relative price distortion. Because $p_t \geq 0.006$ in every regime, the denominator is bounded away from zero in every simulated cell. Event-level price-success differences compare mean price on attributed successes against attributed failures; provider-rank distortion correlates

simulated price ranks with outcome ranks. Table 2 summarizes the regimes and the four probability layers by rule; Table S7 in Online Resource 1 reports convergence diagnostics and Table S8 the full per-layer detail.

Table 2. Generic rare-event simulation: probability layers and pricing distortion by regime and observation rule.

Regime	Observation rule	p objective	p ref observed	p experienced	selection gap	selection ratio	B abs	B rel	B subj bayesian	B price D reputation	D rel prob D reputation	D rel rep D reputation	D rel D reputation	price success diff D reputation	reputation hazard corr	provider rank distortion	prob share D reputation	rep share D reputation
common	endogenous	0.305	0.521	0.521	0.216	1.713	-0.000	-0.001	0.206	0.191	1.027	0.097	1.123	0.119	0.090	-0.357	0.914	0.086
common	exogenous	0.306	0.306	0.305	-0.000	0.999	-0.000	-0.001	-0.001	0.063	0.428	0.207	0.634	0.138	0.120	0.008	0.674	0.326
moderately_rare	endogenous	0.102	0.174	0.174	0.072	1.711	-0.000	-0.001	0.068	0.160	2.072	0.239	2.312	0.032	-0.107	-0.159	0.897	0.103
moderately_rare	exogenous	0.102	0.102	0.102	0.000	1.000	-0.000	-0.003	-0.000	0.115	1.352	0.549	1.901	0.057	0.045	0.019	0.711	0.289
rare	endogenous	0.031	0.052	0.052	0.022	1.710	-0.000	-0.006	0.020	0.124	4.666	0.737	5.403	0.013	-0.159	-0.089	0.864	0.136

Regime	Observation rule	p objective	p ref observed	p experienced	selection gap	selection ratio	B abs	B rel	B subj bayesian	B price D reputation	D rel prob D reputation	D rel rep D reputation	D rel D reputation	price success diff D reputation	reputation hazard corr	provider rank distortion	prob share D reputation	rep share D reputation
rare	exogenous	0.031	0.031	0.031	-0.000	1.000	-0.000	-0.001	-0.000	0.108	3.497	1.752	5.249	0.028	0.014	0.006	0.666	0.334
very_rare	endogenous	0.010	0.017	0.017	0.007	1.708	-0.000	-0.004	0.007	0.094	9.455	2.157	11.612	0.004	-0.171	-0.067	0.814	0.186
very_rare	exogenous	0.010	0.010	0.010	-0.000	1.000	0.000	0.013	0.000	0.089	7.392	5.187	12.579	0.013	0.006	-0.005	0.588	0.412

5. Zero-effect environmental benchmark

The benchmark environment instantiates the framework where efficacy is known exactly: rainmaking—payment for interventions intended to produce rainfall. The benchmark imposes a zero causal effect of rainmaking on rainfall by construction, so every causal efficacy attribution is false by construction (apparent successes are still observed events; what is false is the attribution to intervention). Any performance pricing that emerges is generated by selection, attribution, and reputation alone. This makes the setting a known-zero-effect environmental benchmark rather than the substantive object of the paper: it serves as a known-zero negative-control benchmark, bounding what the mechanisms can produce in the absence of any true effect. The benchmark has two layers that must not be conflated. The OBSERVED-WEATHER layer—actual rainfall, fictional event timings, baseline-hazard selection, and the resulting

apparent-success rates—is an empirical demonstration: real weather shows that selection/timing can generate large apparent-success differences under known zero causal efficacy. The MODEL-OVERLAY layer—willingness to pay, quotes, reputation, and selective memory—remains calibrated and simulated: it demonstrates that the same mechanisms produce the stock–flow pattern when overlaid on real weather, but it measures neither observed market prices, actual willingness to pay, actual reputation formation, nor human probability weighting. The empirical strength of the benchmark is the first layer.

The environmental data are daily precipitation series from the ERA5 reanalysis (Hersbach et al., 2019), comprising 87,664 region-days across 80 climate clusters; Table 3 reports the study locations and coverage. Prospective hazards are estimated with standard baseline-hazard methods (Cox, 1972) on a 1991–2010 training period, and the calibrated willingness-to-pay range follows environmental-valuation and stated-choice practice (Pattanayak & Kramer, 2001; Hensher et al., 2006). Rainmaking institutions are cross-culturally documented, which supplies institutional plausibility for the pricing scenarios without committing the paper to any claim about observed historical fees: ethnographic and historical accounts describe specialized roles, stated payments, and responses to unreliable rainfall (Akong'a, 1987; Espín-Sánchez & Gil-Guirado, 2022; Espín-Sánchez et al., 2026). Prices and acceptance strategies are calibrated scenarios, not reconstructed history.

2595 synthetic pseudo-events are generated under five attribution rules—random dates in qualifying drought conditions, drought dates, preceding dry spells, rolling precipitation deficit, and high-prospective-hazard dates—plus calendar-matched placebos. Figure 2 shows the resulting apparent success rates for the 7-day attribution window at the 1 mm threshold: apparent success differs across event rules purely because baseline hazard differs. The study locations and

coverage are reported in Online Resource 1 (Table S9); Figure 3 maps the empirical hazard structure.

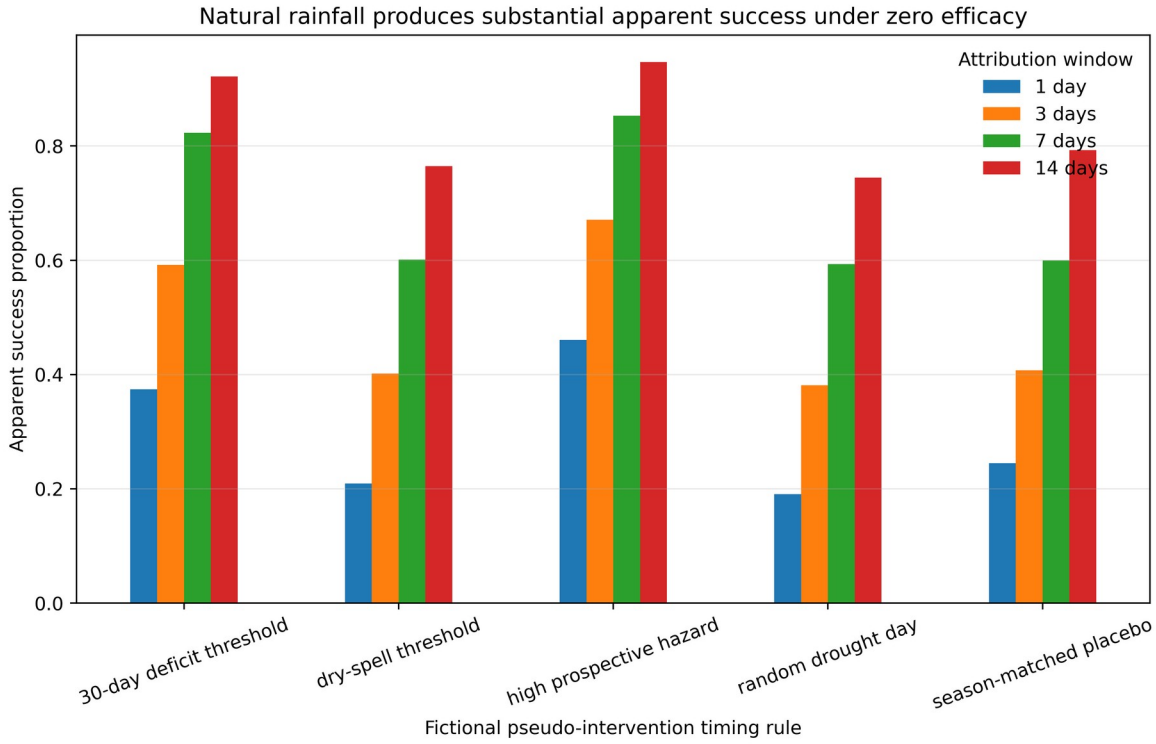


Fig. 2 Natural rainfall following fictional known-zero-effect stress tests in 2011–2020, shown as apparent success by prospective event-selection rule. Apparent successes are observed rainfall events; causal attribution to intervention is false by construction

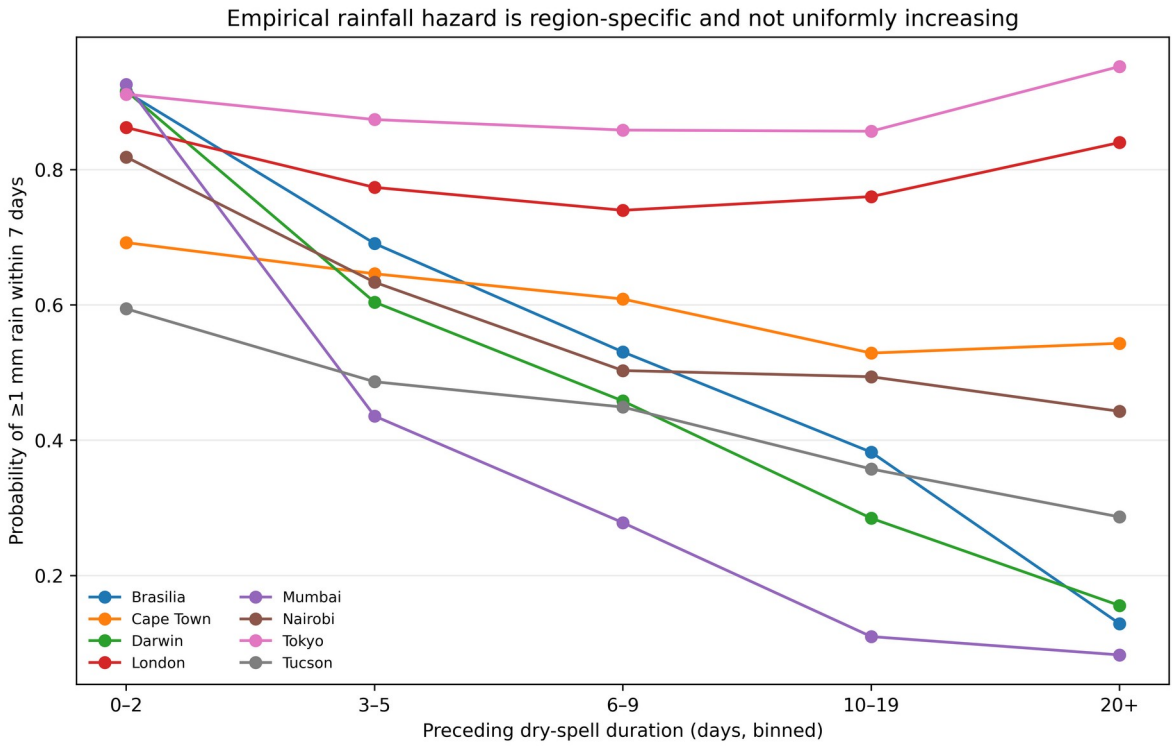


Fig. 3 Estimated seven-day rainfall hazards by preceding dry-spell duration and location in the 1991–2010 training period. The eight locations are pre-specified mechanism stress tests, not a representative global sample

The benchmark Monte Carlo mirrors the generic architecture. Six pricing rules span fixed pricing, naive demand, hazard-adjusted demand, and strategic acceptance with and without reputation-mediated pricing and probability weighting. Because efficacy is exactly zero by construction, observed differences in apparent performance arise only from baseline hazard, selection, attribution, and learning—the precise mechanisms the generic simulation parameterizes.

6. Results

6.1. Risk-correlated selection shifts experienced frequency proportionally, not increasingly with rarity

We decompose the experienced-frequency gap into a matched composition term and a record term: $p_{exp} - p_{obj} = (p_{ref} - p_{obj}) + (p_{exp} - p_{ref})$, where $p_{ref} = E[p_t | O_t = 1]$ is the objective probability on the observed support (Figure 4, panel B; Figure 5, panel A). Under endogenous observation the absolute selection gap falls with rarity—0.216 under common events, 0.072 under moderately rare events, 0.022 under rare events, and 0.007 under very rare events—while the selection ratio is stable at approximately 1.71 to 1.71. This stability is not coincidental: Lemma 1 shows the ratio is exactly independent of baseline rarity under the model's multiplicative hazard and state-dependent selection, and its analytic value 1.7115—the positive-covariance calibration of the general identity—matches the simulated means within Monte Carlo error. The record term $p_{exp} - p_{ref}$ is approximately zero in every regime (0.000 under exogenous observation in the rare regime), so the experienced-frequency inflation is a composition effect: observation selects higher-hazard events, but the record itself is unbiased on the observed support. The earlier formulation of this project—"experienced-frequency bias grows monotonically with rarity"—is therefore corrected: rarity amplifies neither the absolute gap nor the selection ratio.

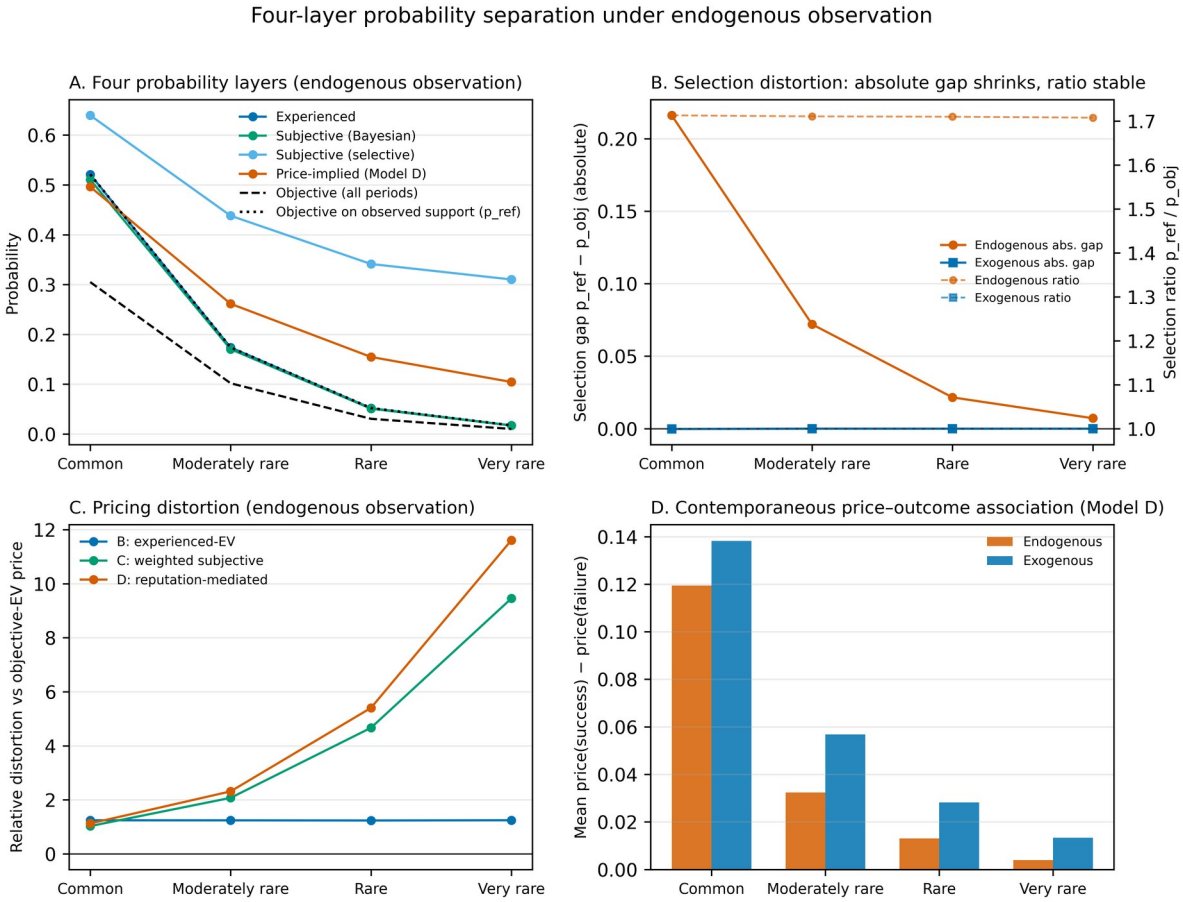


Fig. 4 Four-layer separation in the generic rare-event simulation. Panel A compares the objective probability, experienced frequency, subjective probability, and price-implied index under endogenous observation across rarity regimes; panel B shows the absolute selection gap p_ref − p_obj and the selection ratio p_ref / p_obj under endogenous versus exogenous observation; panel C shows relative price distortion against objective expected-value pricing for models B–D; panel D shows the contemporaneous price–outcome association under Model D. All quantities are generated (simulation evidence, not empirical estimates)

6.2. The four layers separate

Figure 4, panel A, displays all four layers by regime. The subjective layer under selective memory sits above experienced frequency (mean estimate 0.341 against an experienced frequency of 0.052 in the rare regime, so B_{subj} relative to the objective probability is 0.311), while Bayesian updating shrinks experienced frequencies toward the prior; recency weighting tracks the experienced layer closely. The price-implied index under reputation-mediated pricing separates from every probability layer: D_{rel} for model D rises from 1.12 under common events to 11.61 under very rare events. Model A has $D_{rel}=0$ by construction; model B, which prices the experienced frequency, stays near 1.24–1.24 with no trend in rarity, as the rarity invariance of the selection ratio implies; model C rises from 1.03 to 9.45, which is exactly the transformed-frequency part of model D. Figure 5, panel B, decomposes this rise into the two non-scaling components of P2c. The transformed-frequency term is quantitatively the larger share in every simulated regime (9.45 of the 11.61 total in the very-rare regime under endogenous observation —81.4\% of the total, down from 91.4\% under common events), while the additive reputation term grows to 2.16 and its share rises with rarity. Under exogenous observation the pattern is the same with a larger reputation share throughout (58.8\% probability share in the very-rare regime). Neither component is bounded: under the assumed Prelec weighting the transformed-frequency relative term itself diverges as $p \to 0$ (Lemma 2), and the reputation term diverges whenever the reputation stock does not vanish proportionally with p.

Two qualifications keep this from being a denominator tautology. First, the absolute deviation D_{EV} under model D stays approximately flat across regimes (0.175 to 0.129 under endogenous observation): the relative ratio rises because the expected-value benchmark in its denominator shrinks, not because the absolute deviation grows. Second, the economically meaningful

statement is about composition: rarity changes which components dominate the price signal relative to the objective-EV benchmark. Rarity therefore changes the relative composition of the price signal through two non-scaling channels—nonlinear probability transformation, quantitatively dominant on the finite grid, and additive reputation, whose share rises with rarity —not through larger frequency bias and not through a larger absolute deviation. Reputation's distinct result is not the size of its share but the stock–flow decoupling of Section 6.3.

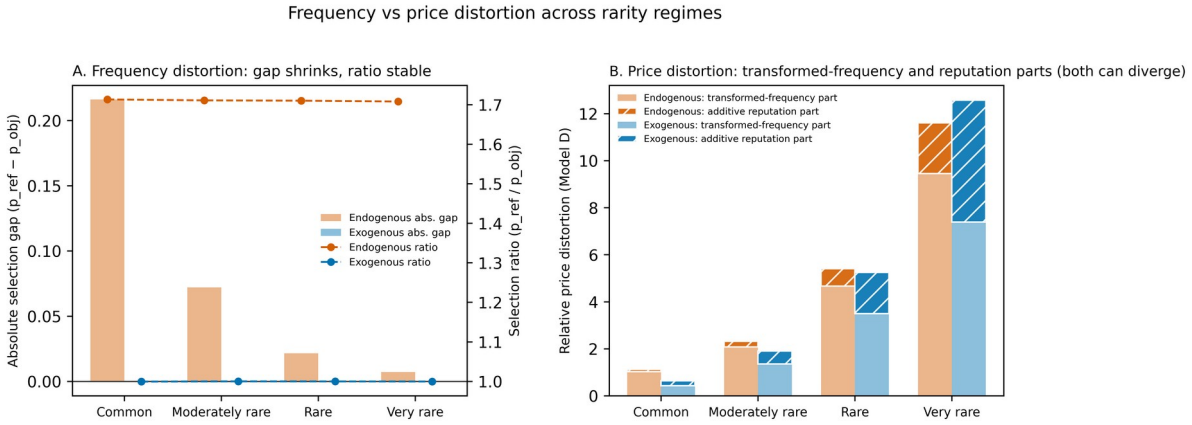


Fig. 5 Frequency distortion versus price distortion across rarity regimes. Panel A: absolute selection gap (bars, left axis) shrinks with rarity while the selection ratio (lines, right axis) is stable at approximately 1.71 under endogenous observation and 1.00 under exogenous observation—the analytically derived invariance of Lemma 1. Panel B: relative price distortion under Model D decomposes into a transformed-frequency part that is quantitatively dominant in every simulated regime and an additive reputation part whose share rises as objective expected value shrinks; neither part is bounded relative to the shrinking benchmark. The relative-distortion rise is a denominator-scaling effect in both non-scaling components, not growing frequency bias and not growing absolute mispricing

6.3. Reputation decouples price from contemporaneous performance

The reputation stock and the contemporaneous outcome flow answer different questions, and the simulation makes the decoupling measurable. The correlation between provider reputation and current hazard is -0.159 in the rare regime—weak and negative, so reputation does not track the current hazard state— while event-level mean price-success differences under model D are small and positive (0.013 in the rare regime) rather than the large positive association a naive performance-pricing account predicts. Provider-rank distortion—the Spearman correlation between price ranks and outcome ranks—is -0.357 under common events, negative because reputation concentrates among providers whose attributed histories pre-date the current hazard cycle. The controlled benchmark below sharpens this result: there, contemporaneous price-success differences are negative.

6.4. Mean-performance equivalence

Figure 6 and Table 3 report the equivalence map, defined on the mean of apparent success only. Under endogenous selection at zero efficacy, mean apparent success in the rare regime matches what an exogenous-observation environment with efficacy of approximately 0.022 would produce (inverse interpolation on the efficacy grid; 0.072 and 0.0075 in the moderately rare and very-rare regimes). In the common regime no efficacy in the scanned grid reproduces the endogenous apparent success—selection alone exceeds what efficacy up to 0.10 explains. The equivalence is first-moment: the matched quantities are mean apparent-success rates, and the joint distribution of outcomes, prices, provider identities, and reputation trajectories is not shown to coincide. A naive observer who regresses outcomes on prices—or compares priced interventions with unpriced periods—recovers a positive "effect" that is entirely generated by

selection. The valid inference is narrower than a price-based identification bound: mean apparent performance alone cannot distinguish zero-efficacy endogenous selection from positive-efficacy exogenous observation in the simulated rare regimes, and observed quote data do not automatically solve the problem because quotes themselves load on transformed beliefs and reputation.

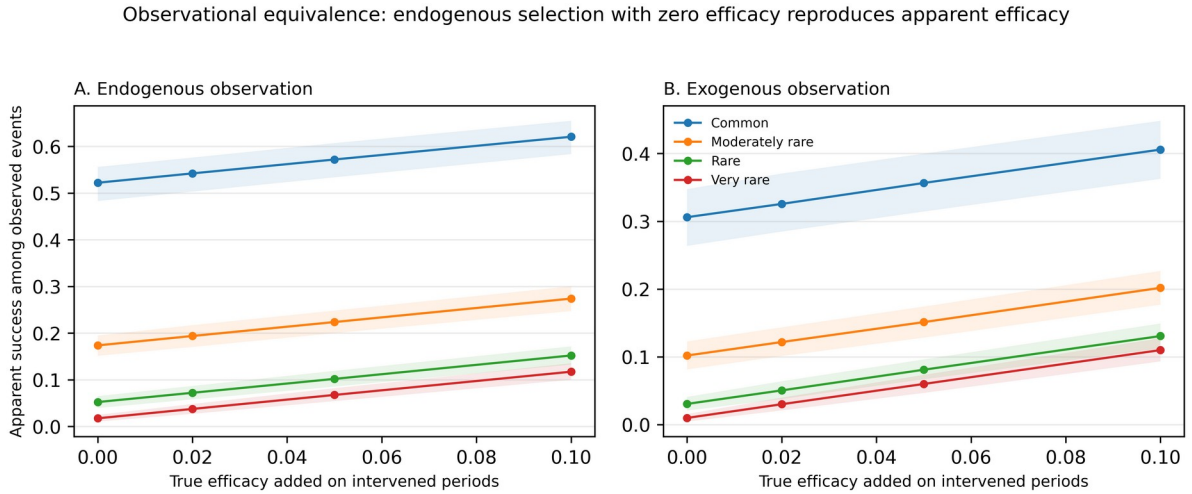


Fig. 6 Mean-performance equivalence map. Mean apparent success among observed events as true efficacy added on intervened periods varies, under endogenous (A) and exogenous (B) observation. Endogenous selection with zero true efficacy reproduces the MEAN apparent success of positive efficacy under exogenous observation (first-moment equivalence only)

Table 3. Mean-performance equivalence: mean apparent success by true efficacy, regime, and observation rule.

Regime	Observation rule	0.0	0.02	0.05	0.1
common	endogenous	0.522	0.542	0.572	0.621
common	exogenous	0.306	0.326	0.357	0.406

Regime	Observation rule	0.0	0.02	0.05	0.1
moderately_rare	endogenous	0.174	0.194	0.224	0.274
moderately_rare	exogenous	0.102	0.122	0.151	0.202
rare	endogenous	0.052	0.072	0.102	0.152
rare	exogenous	0.031	0.051	0.081	0.131
very_rare	endogenous	0.018	0.038	0.068	0.117
very_rare	exogenous	0.010	0.030	0.060	0.110

6.5. Benchmark: apparent performance under exact zero efficacy

The environmental benchmark reproduces the generic mechanisms at extreme clarity. Apparent success reaches 0.853 for high-prospective-hazard events—versus 0.600 for matched placebos and 0.593 for random dates—under exactly zero efficacy (Figure 2). These apparent successes are observed rainfall events; what is false by construction is their causal attribution to the intervention.

Overlaying the calibrated pricing and reputation rules on the same weather benchmark reproduces the stock–flow pattern of Section 6.3 in sharpened form: reputation-mediated premia persist while mean contemporaneous price–success differences are negative. This overlay is a model result, not observed pricing—the empirical content of the benchmark is the selection/apparent-success demonstration above. The scenario WTP regressions, permutation falsification, and M4/M5 quote-reputation decomposition are reported in Online Resource 1 (Section S15); Table 4 reports the pseudo-event summary. Figure 7 shows the full Monte Carlo decomposition (detail in Online Resource 1, Table S10), and Figure 8 reports robustness.

Table 4. Zero-effect benchmark: pseudo-event apparent success.

Selection rule	Events	Apparent success 1d	Apparent success 3d	Apparent success 7d	Apparent success 14d	Mean scenario WTP
30-day deficit threshold	294	0.374	0.592	0.823	0.922	6.276
dry-spell threshold	727	0.209	0.402	0.601	0.765	3.267
high prospective hazard	395	0.461	0.671	0.853	0.947	6.815
random drought day	509	0.191	0.381	0.593	0.745	3.073
season-matched placebo	670	0.245	0.407	0.600	0.793	3.550

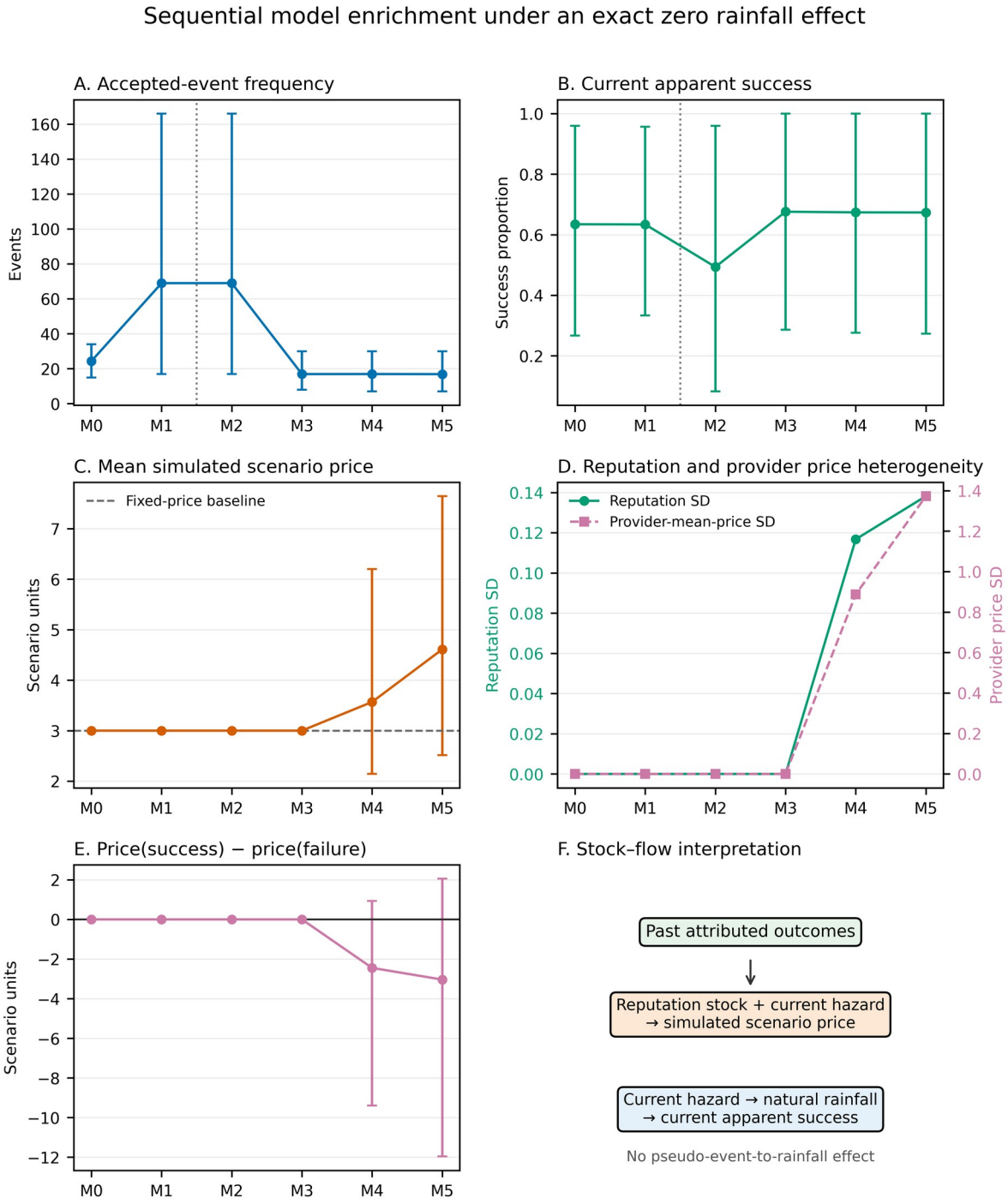


Fig. 7 Zero-effect benchmark: sequential Monte Carlo pricing-rule enrichment under M0–M5 on fictional rainmaking pseudo-events over observed weather. M0–M1 shuffle rainfall outcomes within sampled years; M2–M5 retain observed local and seasonal ordering.

Selective acceptance changes accepted-event composition and current apparent success. In M4–M5, past attributed outcomes accumulate in reputation stocks that create price premia and provider-price heterogeneity, while current success remains a natural-rainfall flow governed by current hazard. Mean price(success) minus price(failure) is negative in M4–M5. Error bars are 2.5th and 97.5th Monte Carlo percentiles, not confidence intervals for historical parameters

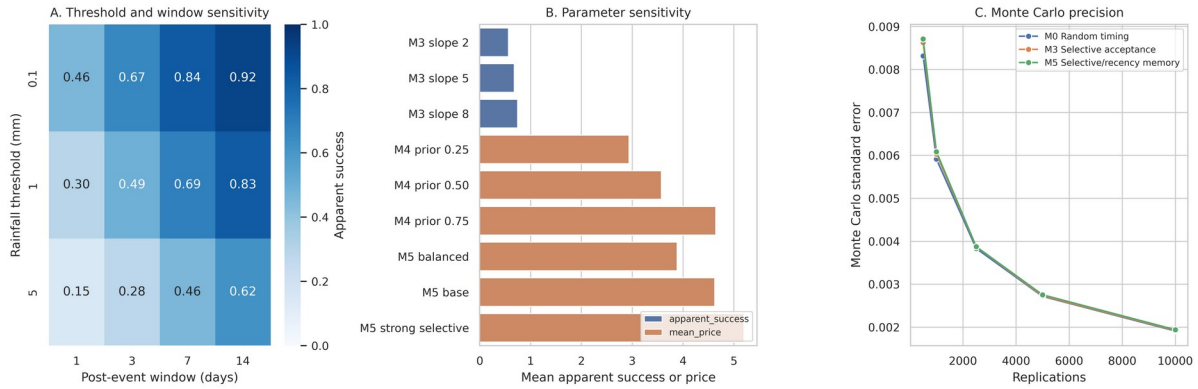


Fig. 8 Robustness. Apparent success across rainfall thresholds and windows, selected parameter sensitivity, and Monte Carlo precision

7. Discussion

7.1. What prices reveal

The framework's main output is a diagnostic: a rare-event price is a composite signal, and the composition changes with rarity. The selection/composition mechanism is already fully present under common events—a selection ratio of approximately 1.71 and a relative price distortion of 1.12 under model D—so rarity does not create the separation. What rarity changes is the scaling and composition of the price signal. Under the positive risk–observation covariance calibration

used here, endogenous selection inflates the experienced layer, transformation pushes the subjective layer away from it in either direction, and reputation loads the price-implied index with a stock that is nearly orthogonal to contemporaneous performance. In the benchmark environment the composite is extreme—prices remain elevated while mean contemporaneous price-success differences are negative—demonstrating that the sign of the price-performance correlation is not a test of whether pricing works.

What is uniquely rare-event about the paper is therefore not the selection mechanism, which is general and operates identically in common regimes (Lemma 1). It is the economics of scale: as objective expected value shrinks, both non-scaling components—nonlinear probability transformation and additive reputation stocks—grow relative to the objective-EV benchmark, successful outcomes become sparser, and the equivalent-efficacy interpretation of a raw apparent-success rate shrinks with the selection gap: the mean performance that zero-efficacy selection produces falls with the baseline probability, so the equivalent efficacy scales downward (0.072 to 0.0075 across the simulated rare regimes) rather than staying inflated. Rarity changes the interpretation of the price, not the existence of the distortion.

7.2. Identification

The mean-performance equivalence result suggests concrete routes to separating selection from efficacy within the present framework: (i) disclosure of the rejected case mix, which would break the observation filter; (ii) an exogenous observation rule, so that experienced frequencies are generated independently of the risk state; (iii) pre-event risk information, which would let the case mix be adjusted directly; or (iv) a known-zero benchmark that calibrates what the mechanisms alone produce. The benchmark environment supplies (iv); the others are design routes for markets that price rare interventions, and other identification strategies may also be

possible. These routes parallel the case-mix adjustments developed for surgeon report cards, where baseline risk and selection contaminate raw performance comparisons (Shahian & Normand, 2008; Werner & Asch, 2005; Barnett et al., 2005), and the information-aggregation concerns of cascade models in which endogenous signals transmit distorted states (Bikhchandani et al., 1992; Dranove et al., 2003; Kreps & Wilson, 1982), and expert advice embeds similar adverse-selection and reputation forces (Liu & Sanyal, 2012).

7.3. Candidate application domains

The framework is a structural analogy, not a validated account of any particular market. It applies as a candidate model only where four ingredients jointly hold: a rare priced outcome, endogenous observation (case selection correlated with the risk state), a hidden counterfactual, and prices or reputations built on the observed record. Medical reputation markets plausibly satisfy all four—low-frequency outcomes, hidden counterfactuals, selected case loads—and surgeon report cards already document the raw-comparison problem the framework formalizes (Shahian & Normand, 2008; Werner & Asch, 2005), although their information structure, regulation, and price-setting institutions differ from the stylized model. The direction of the composition shift is domain-specific: by Lemma 1 it is determined by $Cov(m, s)$, so provider cherry-picking of low-risk cases can inflate apparent success while adverse selection can move it in the opposite direction. Catastrophe insurance, credit for rare default events, and low-frequency professional services are candidate domains to the extent that providers or buyers select endogenously on risk and histories accumulate into reputation—experienced price distributions demonstrably shape subsequent consumer beliefs (Matsumoto & Spence, 2016); whether each ingredient holds—and which sign the covariance takes—is an empirical question this paper does not answer. Where the ingredients do hold, the framework predicts (a) experienced-frequency

inflation under positive risk–observation covariance (deflation under negative), concentrated in the observation-selected segment, (b) price-implied indices that track reputation stocks rather than contemporaneous flows, and (c) attenuation of apparent efficacy once hazard is controlled —the pattern the benchmark environment exhibits in its extreme form.

7.4. Limitations

Three limitations bound the claims. The simulation demonstrates what the mechanisms can produce, not what actual people or markets do; no claim is made about observed historical rainmaking prices, universal probability weighting, actual consumer willingness-to-pay relationships, or observed market pricing of rare events. The generic model is stylized: one risk state, stationary rarity regimes, and specific functional forms for demand, acceptance, and learning, although sensitivity checks vary the key slopes. The benchmark uses fictional pseudo-events over real weather; it demonstrates the feasibility of the mechanisms under a true zero but measures neither market prices nor human beliefs in any real domain. The quote rule is one reduced-form mapping—other pricing institutions could shift magnitudes though not the decomposition itself—and the reputation stock follows exponential smoothing of attributed outcomes, one accumulation rule among others; the stock–flow decoupling it demonstrates is a possibility result, not a claim about every reputation mechanism.

8. Conclusion

When the event being priced is rare, the counterfactual hidden, and observation endogenous, the signal embedded in a modeled quote is not generally the objective risk. The framework separates the objective probability, the experienced frequency on the selected record, the transformed subjective probability, and a price-implied index that need not be a probability at all; under the

positive risk–observation covariance calibration used here, endogenous selection inflates experienced frequencies by a selection ratio that is analytically independent of baseline rarity under multiplicative scaling, while reputation-mediated pricing decouples price from contemporaneous performance and produces a relative distortion that rises with rarity through two non-scaling channels—a nonlinear probability-transformation term that carries the larger share in the simulated regimes and an additive reputation term whose share grows—an absolute deviation that stays flat even as the relative ratio grows; and the zero-effect environmental benchmark demonstrates that elevated apparent performance, reputation premia, and negative mean contemporaneous price-success differences all arise under exactly zero efficacy. Prices of rare events track a composite of baseline hazard, selection, transformation, and accumulated attribution—and observed model-implied quotes do not by themselves reveal how much of the signal reflects each component. Within the present framework, evaluation of rare-event pricing can be aided by case-mix disclosure, exogenous observation, pre-event risk information, or known-zero benchmarks—several routes rather than an exclusive requirement; the framework supplies the language, and the benchmark supplies the zero effect.

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Statements and Declarations

Data and code availability

Code, frozen analytical outputs, acquisition ledgers, and manuscript-generation files are publicly available at https://github.com/bougtoir/pricing-the-rain. The ERA5 precipitation data

(Copernicus Climate Change Service) are not redistributed. They are publicly available through the Open-Meteo Historical Weather API, and the repository provides the exact request URLs, retrieval code, and checksums of the derived files. Other sources whose redistribution rights are unclear are likewise not redistributed. Fictional pseudo-events and all simulation outputs are reproducible from the re-acquired rainfall series and the declared seeds.

Funding

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Competing interests

[AUTHOR INPUT REQUIRED: declare competing interests or state that none exist.]

Author contributions

[AUTHOR INPUT REQUIRED: provide CRediT roles for each author.]

Ethics

The study uses public environmental data, bibliographic metadata, and simulated fictional events. It includes no human participants, personal data, or animal research.

Generative AI disclosure

During preparation of this work, the authors used Devin (Cognition AI) to assist with code development, manuscript drafting, and language organization. [AUTHOR CONFIRMATION REQUIRED: after using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.] No generative-AI image tool was used to

create or alter the scientific figures, and no generative AI was used to fabricate data or references.

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